



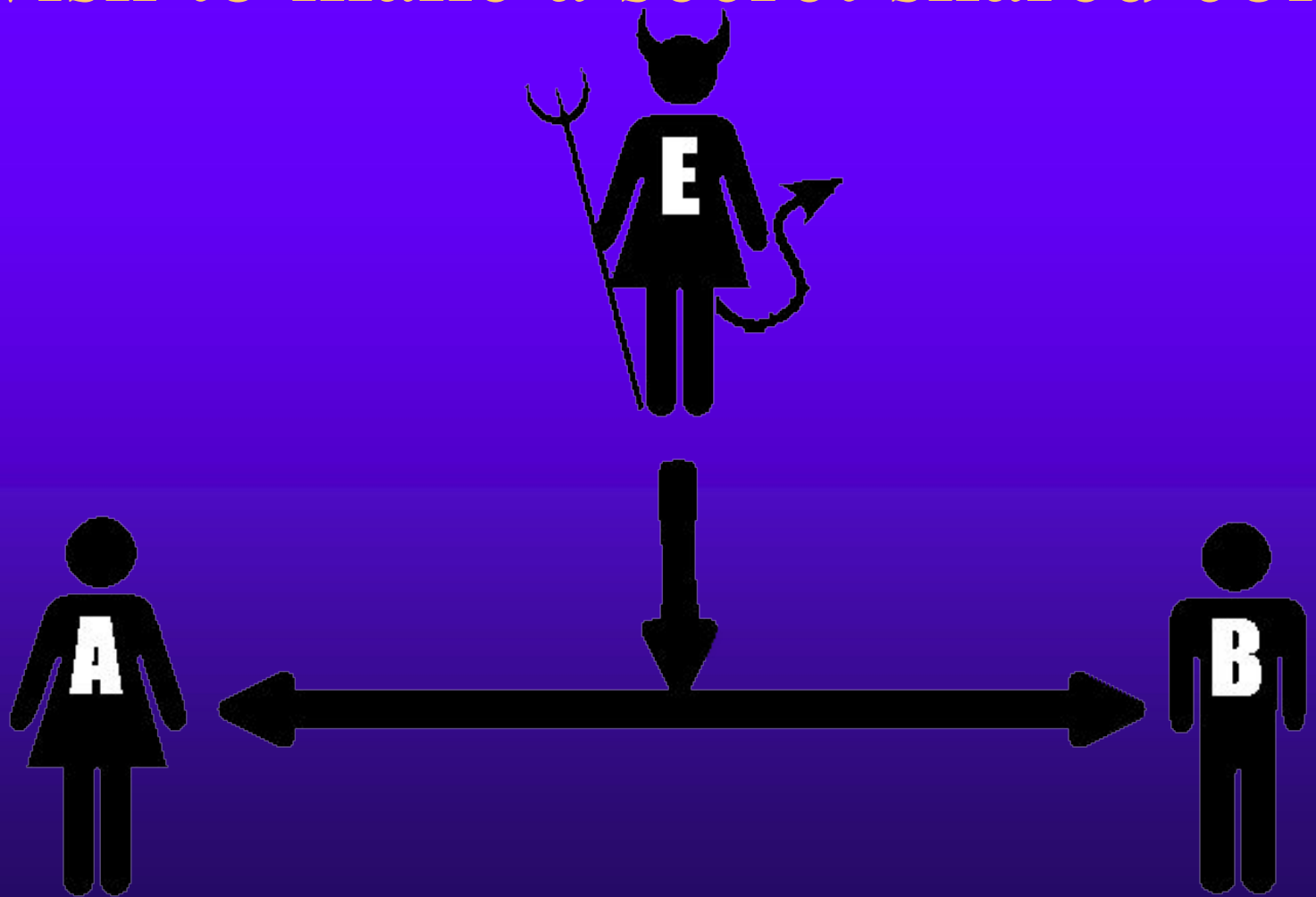
Diffie-Hellman Key Exchange



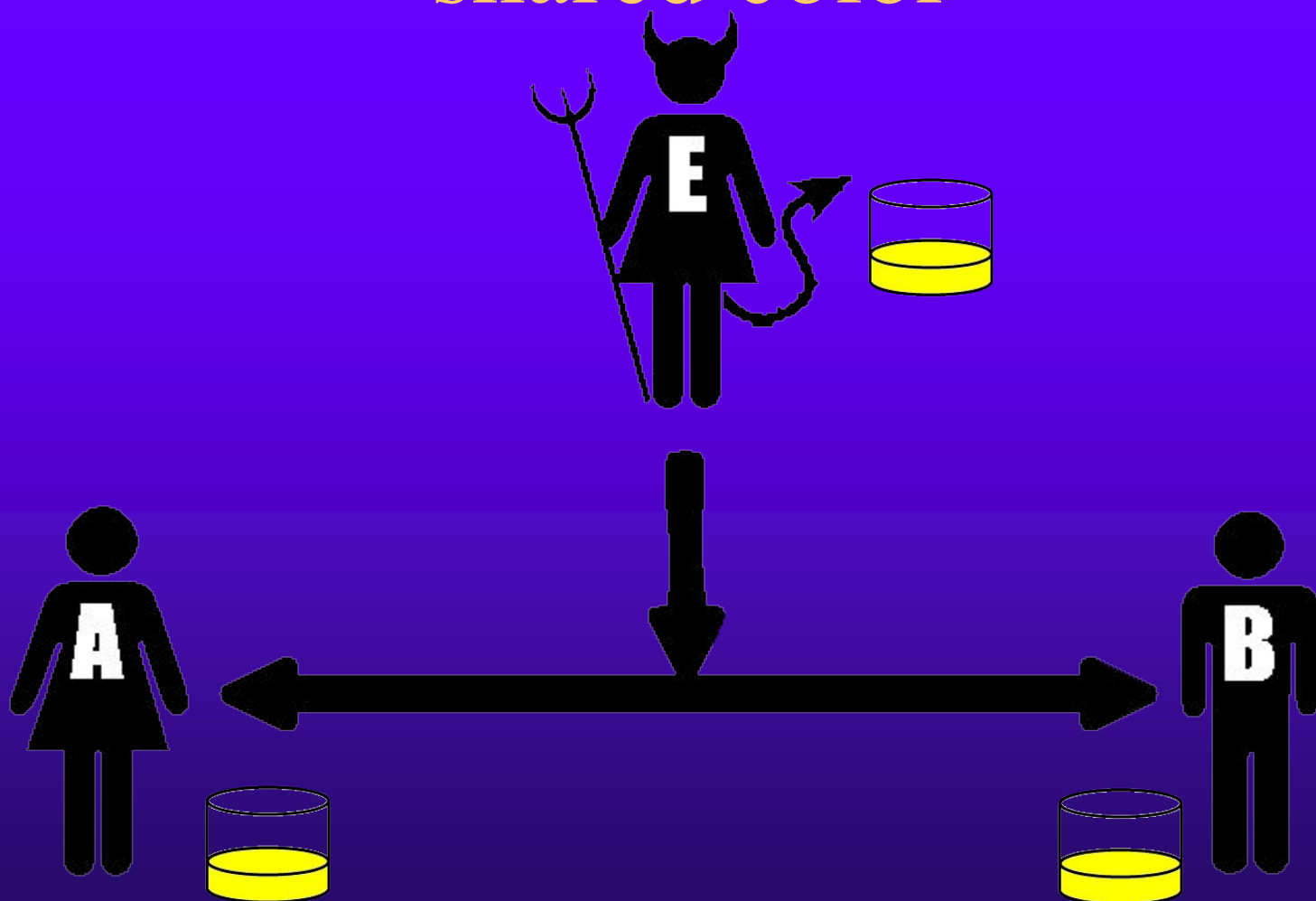
The Problem of Key Exchange

- One of the main problems of symmetric key encryption is it requires a secure & reliable channel for the shared key exchange.
- The Diffie-Hellman Key Exchange protocol offers a way in which a public channel can be used to create a confidential shared key.

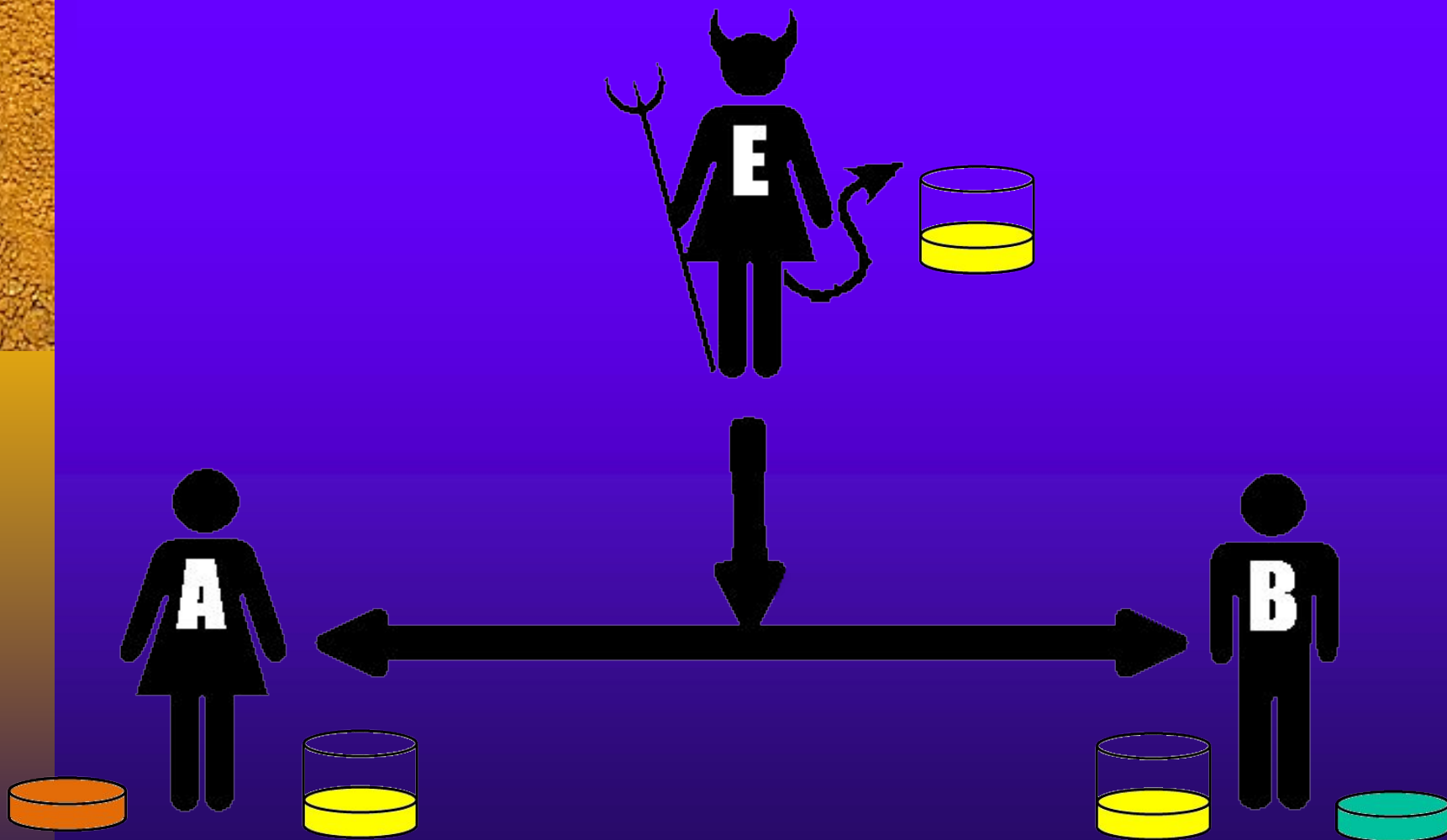
Alice & Bob with Eve listening
wish to make a secret shared color



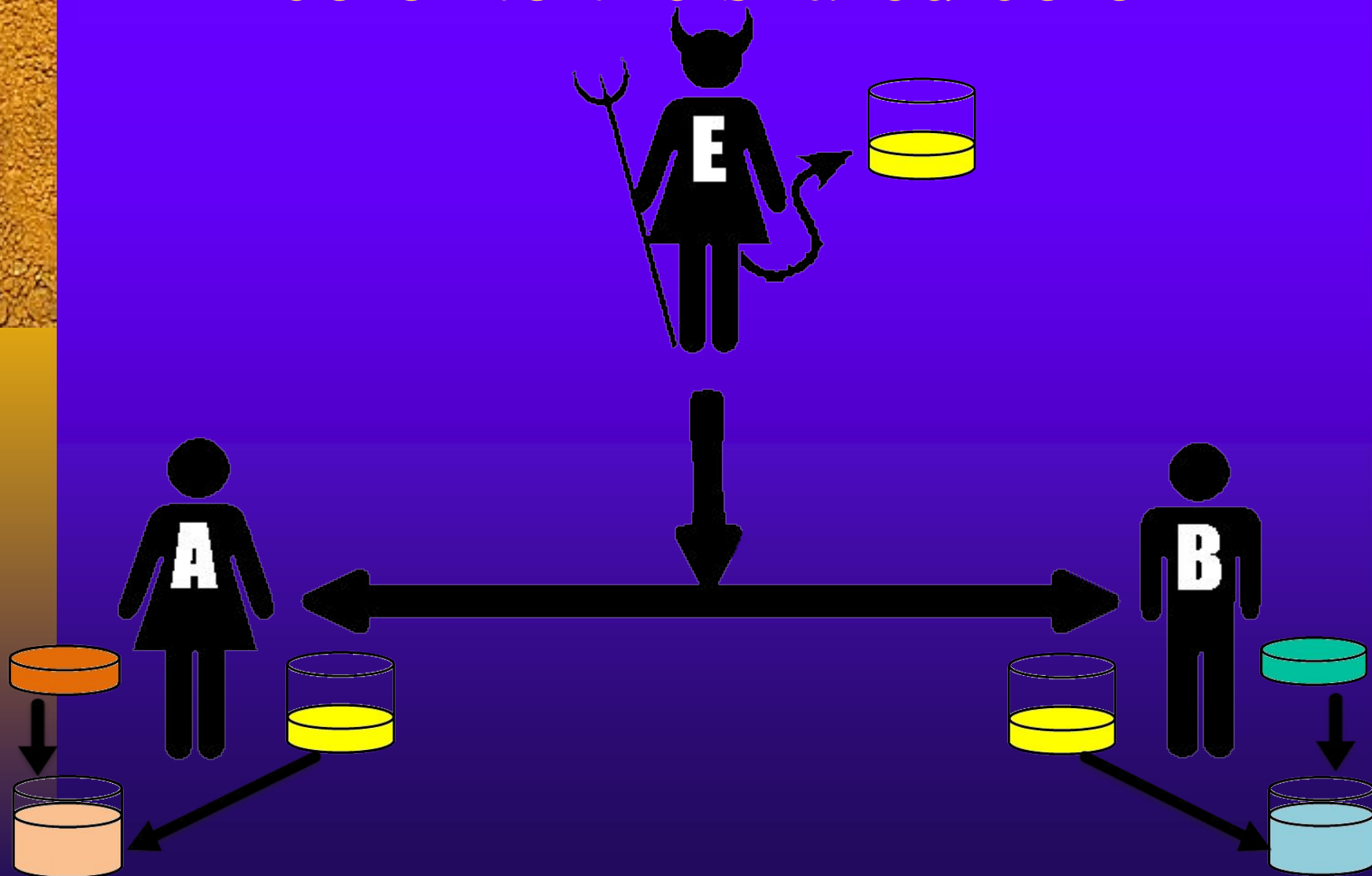
Step 1 - Both publicly agree to a shared color



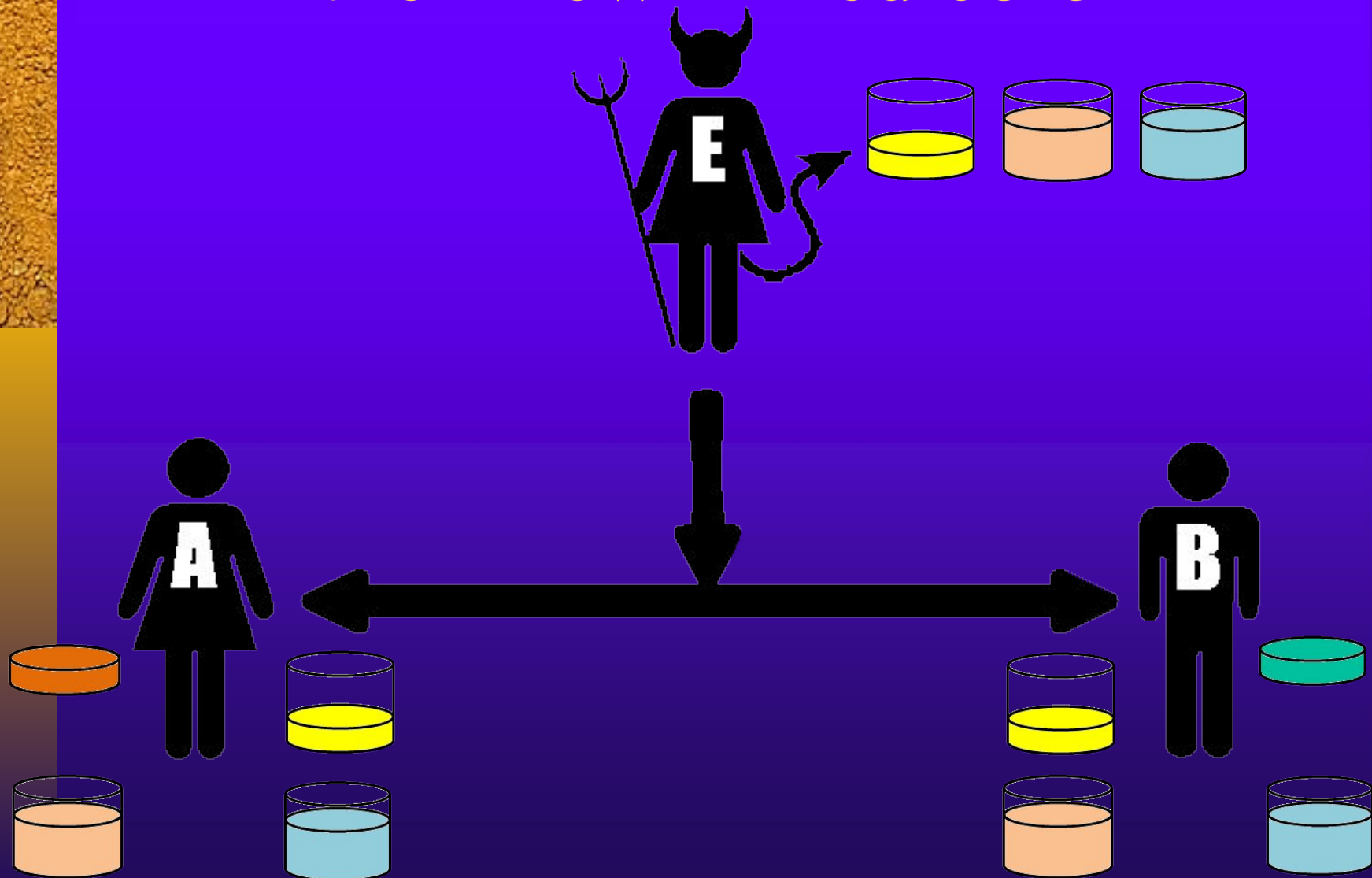
Step 2 - Each picks a secret color



Step 3 - Each adds their secret color to the shared color

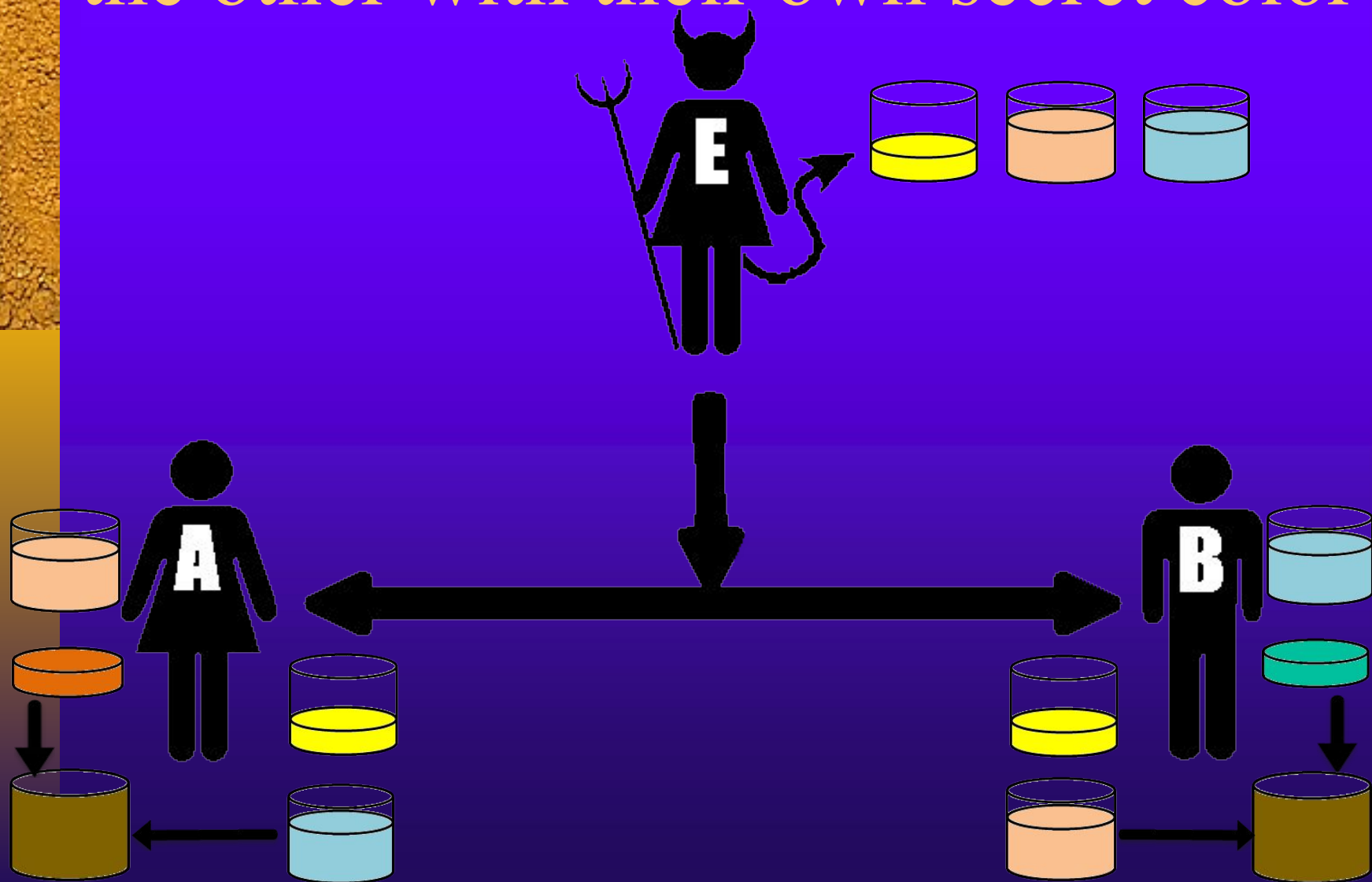


Step 4 - Each sends the other
their new mixed color





Each combines the shared color from the other with their own secret color





Alice & Bob have agreed to a shared color unknown to Eve

- How is it that Alice & Bob's final mixtures are identical?
- Alice mixed
 - $[(\text{Yellow} + \text{Teal})_{\text{from Bob}}] + \text{Orange}$
- Bob mixed
 - $[(\text{Yellow} + \text{Orange})_{\text{from Alice}}] + \text{Teal}$



Alice & Bob have agreed to a shared color unknown to Eve

- How is it that Alice & Bob's final mixture is secret?
- Eve never has knowledge of the secret colors of either Alice or Bob
- Unmixing a color into its component colors is a hard problem



Alice & Bob have agreed to a shared color unknown to Eve

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Diffie-Hellman Key Exchange

Adding Mathematics

Diffie-Hellman Key Exchange Agreement/Algorithm

Press Esc to exit full screen

Diffie-Hellman Key Exchange/Agreement Algorithm

- >> Two parties, can agree on a symmetric key using this technique.
- >> This can then be used for encryption/ decryption.
- >> This algorithm can be used only for key agreement, but not for encryption or decryption.
- >> It is based on mathematical principles.

Algorithm -

1. Firstly Alice & Bob agree upon 2 large prime numbers - n & g
These 2 numbers need not be secret & can be shared publicly.
2. Alice chooses another large random number x (private to her) & calculates A such that : $A = g^x \bmod n$
3. Alice sends this to Bob.
4. Bob chooses another large random number y (private to him) & calculates B such that : $B = g^y \bmod n$
5. Bob sends this to Alice.
6. Alice now computes her secret key $K1$ as follows:
 $K1 = B^x \bmod n$
7. Bob computes his secret key $K2$ as follows:
 $K2 = A^y \bmod n$
8. $K1 = K2$ (key exchange complete)

